UM-1: a 4+4 model of the Universe with Einstein-Lovelock gravity, the dirac16complex field and an x0-generated mass spectrum

dirac16complex project (with Claude Opus 5.5)

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Version 1.1 of 2026-10-17: version 1 plus the two changes that the solution forces (marked CHANGED in sections 4.1 and 8; the reasons are in `SOLUTION.md` section 3). Written at the author's order of 2026-10-17 ("create a new model of the Universe"). This document DEFINES the model. Its field equations are derived in `FIELD_EQUATIONS.md` (scripts in `field_equations/`), an approximate solution is computed in `SOLUTION.md` (scripts in `solution/`), and the origin of every ingredient is in `PROVENANCE.md`. Every statement below is labelled as one of: DERIVED (computed in this repository, with the file named), TAKEN (from the Revision record, file named), STANDARD (textbook physics, used as input), ASSUMED (a choice of this model), INVENTED (a new phenomenological ingredient of this model) or OPEN.

1. Summary

The model lives on an eight-dimensional space with the author's coordinates $x_0, x_1, \ldots, x_7$: x_0 is the hidden space-like direction, x_1, x_2, x_3 are ordinary space, x_4 is the time and x_5, x_6, x_7 are three extra time-like directions that deflate exponentially. Gravity is Einstein-Lovelock gravity in eight dimensions with the author's couplings. The dirac16complex field Ψ, a 16-component spinor of $Cl(4,4)$ coupled to gravity through the canonical spin connection, is the dark-matter candidate. Photons, neutrinos, electrons, up and down quarks, the gluons (required for consistency with QCD), and the $W^{\pm}$ and Z bosons are eight-dimensional fields of the gauge group $SU(3)_c \times SU(2)_L \times U(1)_Y$. There is NO Higgs field and no Higgs particle.

The central new idea (the author's): every field is a function of x_0 that is periodic with the period $\ell_0 = \pi/(6H)$ of the metric. Expanding a field in the eigenfunctions of its x_0 operator turns the eight-dimensional wave equation into a tower of four-dimensional fields whose masses are the eigenvalues. A mode that does not depend on x_0 is massless (graviton, photon, gluon). A mode localised in x_0 carries a hidden momentum $\Delta p_{x_0} \geq \hbar/(2\,\Delta x_0)$ and therefore an effective mass. The model makes this precise: the four-dimensional mass of every massive particle is the lowest eigenvalue of its x_0 operator on the periodic cell, with the boundary conditions at the two special points of the cell (the fold $z = \pi/2$ and the tip $z = 0$), and the uncertainty relation is a lower bound that every such eigenvalue satisfies.

2. Space, coordinates and the periodic cell

- Coordinates (TAKEN, `Revision/SPEC.md` section 1): $z = 6Hx_0$ takes every real value; the metric depends on x_0 only through $\cot^2 z$ and $(\sin^2 z)^{1/6}$, both periodic with period π in z, i.e. with period $\ell_0 = \pi/(6H)$ in x_0.
- The cell (TAKEN, `Revision/kohn_sham/ks-theory.json`, boundaryConditions): one period $0 < z < \pi$ consists of two halves, $0 < z < \pi/2$ and $\pi/2 < z < \pi$, joined at the FOLD $z = \pi/2$ ($\cos z = 0$, where g_{00} and $\sqrt{|g|}$ vanish) and ending at the TIP $z \to 0, \pi$ ($\sin z \to 0$). The hidden coordinate $y = \ln|\sin z|/(6H) \in (-\infty, 0]$ measures the distance from the fold; $dy = \cot z\, dx_0$.
- A field periodic in x_0 with period ℓ_0 is a field on the cell. Its two halves are related by the fold condition $(1 - Q(\theta))\chi(0) = 0$, $Q(\theta) = \cos\theta\,\sigma_3 + \sin\theta\,\sigma_2$, with $\theta = 0$ (even) or $\theta = \pi$ (odd) (DERIVED in the Kohn-Sham record up to θ). Periodicity in x_0 requires that the two tips of a cell ($z \to 0^+$ and $z \to \pi^-$) be identified; the proper distance to the tip is infinite, so

the model replaces the tip by a regular cutoff at $y = -L$ (ASSUMED, as in the Kohn-Sham record; `tip_convergence/` measures the dependence on L).

3. The metric as a power series

The metric is written as a series of terms of decreasing magnitude (ASSUMED structure):

$$g_{MN} = \bar{g}_{MN} + \epsilon\, h^{(1)}_{MN} + \epsilon^2 h^{(2)}_{MN} + \dots$$

- Order 0 (the background, TAKEN from `Revision/SPEC.md` section 1, the corrected primordial metric, version 2):

$$\bar{g} = \mathrm{diag}(e^{-2a_4} b_4 \cot^2 z,\ e^{2a_4},\ e^{2a_4},\ e^{2a_4},\ -1,\ -e^{-2a_4} b_4 (\sin^2 z)^{1/6},\ -e^{-2a_4} b_4 (\sin^2 z)^{1/6},\ -e^{-2a_4} b_4 (\sin^2 z)^{1/6})$$

with two GENERAL functions of the time, $a_4(x_4)$ (the observer's scale factor $a = e^{a_4}$) and $b_4(x_4) > 0$. The block (x_0, x_5, x_6, x_7) has the common scale factor $f = e^{-a_4} b_4^{1/2}$ and the rate $h_4 = \dot{f}/f = u/2 - \dot{a}_4$, $u = \dot{b}_4/b_4$; the extra times deflate when $h_4 < 0$. The primordial history of the record is $a_4 = H x_4$, $b_4 = e^{(4/3) H x_4}$ ($h_4 = -H/3$). The model keeps a_4 and b_4 general and determines them from the field equations with all sources (section 7).

- Order 1: the cosmological perturbations, in the 3+1 decomposition of $(x_1, \dots, x_4)$: the scalar potentials Φ, Ψ_N (Newtonian gauge, $\delta g_{44} = -2\Phi$, $\delta g_{ij} = -2\Psi_N e^{2a_4} \delta_{ij}$), the tensor modes h^{TT}_{ij}, and the block perturbation $\delta b_4/b_4 = 2\sigma$ (a radion), all functions of $(x_1, \dots, x_4)$ times the flat x_0 zero-mode profile; the x_0-dependent parts of the sources drive x_0-dependent corrections, which are expanded in the same x_0 eigenfunctions as the fields and are heavy (section 5). The bookkeeping parameter ϵ is of the size of the CMB temperature anisotropy, $\epsilon \sim 10^{-5}$ (STANDARD).
- Order 2 and beyond: the products of order-1 quantities and the x_0-heavy modes; not computed in version 1 (OPEN).

4. Fields and action

$$S = S_{EL}[g] + S_\Psi[g, \Psi] + S_{SM}[g, A, \psi] + S_{tip}[g, A, \psi, \Sigma] + S_{portal}[\Psi, A, \psi]$$

4.1 Einstein-Lovelock gravity (TAKEN)

$$S_{EL} = \frac{1}{2\kappa} \int d^8x \sqrt{|g|} \Big(\sum_{k=1}^{3} \alpha_k\, \mathcal{L}_{(k)} - 2\Lambda \Big), \qquad \sum_k \alpha_k E_{(k)}{}^M{}_N + \Lambda\, \delta^M_N = \kappa\, T^M{}_N$$

with $\mathcal{L}_{(1)} = R$, $\mathcal{L}_{(2)}$ the Gauss-Bonnet and $\mathcal{L}_{(3)}$ the cubic Lovelock density, the author's weights $\alpha_k = w_k H^{2-2k}$, $w = (1, 1/32, 1/1152)$, and $\Lambda = -10H^2$ in the convention of `Revision/field_equations_a4/a4-equations.json`. The graviton is the x_0 zero mode of g_{MN} and is massless.

CHANGED in version 1.1 (`SOLUTION.md` section 3, DERIVED there): with the author's weights the only static vacuum, $k_0 = 1/(4w_2) = 8$ with $\Lambda_0 = -24H^2$, has the effective-Newton factor $G(k_0) = -5 + 9w_3/(2w_2^2) = -1$, i.e. repulsive four-dimensional gravity, and $\Lambda = -10H^2$ admits no static block with a small cosmological term. Version 1.1 uses $w = (1, 1/32, 1/768)$ ($G(k_0) = 1$, the Einstein value) and $\Lambda = -24H^2 + \delta\Lambda$, and ASSUMES a block stabiliser (an extra potential for $k = e^{2a_4}/b_4$ with curvature above $k_0 G(k_0) H^2/4 = 2H^2$, since the block mode of the homogeneous reduction has $\lambda^2 = k_0 G(k_0) H^2/4$: attractive gravity makes the static block unstable). The origin of the stabiliser (the Casimir energy of the cell) is OPEN.

4.2 The dirac16complex field (TAKEN)

$$S_\Psi = \int d^8x \sqrt{|g|}\left[\tfrac{1}{2}(\bar\Psi\gamma^M D_M\Psi - (D_M\bar\Psi)\gamma^M\Psi) - m\,\bar\Psi\Psi - \tfrac{\lambda}{2}(\bar\Psi\Psi)^2\right], \quad D_M = \partial_M + \Omega_M,\ \Omega_M = \tfrac{1}{2}\omega_M{}^{ab} S_{ab}$$

with $\bar\Psi = \Psi^\dagger C$, the author's γ matrices (`Revision/algebra/gammas.json`) and the CANONICAL spin connection of the diagonal vielbein of $\bar g$, which is non-trivial: $\gamma^M\Omega_M = \frac{3}{2}He^{a_4}b_4^{-1/2}\gamma^{x_0} + (u - \frac{1}{2}\dot a_4)\gamma^{x_4}$ (DERIVED, `Revision/lead_checks/`). Ψ is a singlet of $SU(3)_c \times SU(2)_L \times U(1)_Y$ (ASSUMED: dark). Its quanta are of three kinds (DERIVED in the Kohn-Sham record): the fold-band quanta, $\varepsilon = e^{-a_4}|k|$ exactly, MASSLESS in four dimensions (the fold zero modes of the x_0 operator); the bulk-band quanta with $\varepsilon \geq m$ (massive; the x_0 momentum adds $(l\pi/(fL))^2$ to m^2); and the homogeneous condensate $S = \bar\Psi\Psi = S_0 e^{a_4} b_4^{-2}$.

4.3 The Standard-Model sector in eight dimensions (ASSUMED structure, STANDARD content)

- Gauge fields G^α_M ($SU(3)_c$), W^a_M ($SU(2)_L$), B_M ($U(1)_Y$) with $S_{gauge} = -\frac{1}{4}\int d^8x\sqrt{|g|}\, g^{MP}g^{NQ}(\frac{1}{\hat g_s^2}G^\alpha_{MN}G^\alpha_{PQ} + \frac{1}{\hat g^2}W^a_{MN}W^a_{PQ} + \frac{1}{\hat g'^2}B_{MN}B_{PQ})$; the four-dimensional couplings are $g^2 = \hat g^2/V_\perp$ etc. with $V_\perp$ the proper volume of the cell and the extra-time block (section 5).
- One generation of fermions, each an eight-dimensional spinor of $Cl(4,4)$ (16 components, the same algebra as Ψ): $L = (\nu, e)_L$ (doublet, $Y = -\frac{1}{2}$), e_R ($Y = -1$), ν_R ($Y = 0$), $Q = (u, d)_L$ (colour triplet, doublet, $Y = \frac{1}{6}$), u_R ($Y = \frac{2}{3}$), d_R ($Y = -\frac{1}{3}$), with $S_\psi = \int d^8x\sqrt{|g|}\,[\bar\psi\gamma^M D_M\psi - \mu_\psi\,\bar\psi\psi]$, D_M containing the same canonical spin connection and the gauge fields. Each species has a bulk mass μ_ψ (constant on the cell) and its own fold condition θ_ψ (INVENTED assignment): exactly as for Ψ, the reduced x_0 operator is the block $h = -(i/f)\sigma_1\partial_y + \mu_\psi\sigma_2$ and, with the regular tip $b(-L) = 0$, the condition $\theta = 0$ leaves the single zero mode $\chi = (e^{f\mu_\psi y}, 0)$ (DERIVED in the Kohn-Sham record), a chiral four-dimensional fermion localised at the fold for $\mu_\psi > 0$ and at the tip for $\mu_\psi < 0$. The doublets L, Q carry the left-handed zero modes, the singlets e_R, ν_R, u_R, d_R the right-handed ones (the opposite γ^{x_0} component, the zero mode $(0, e^{-f\mu_\psi y})$ with the other fold condition $a(0) = 0$ and the transformed tip condition $a(-L) = 0$: the map $\chi \to \sigma_2\chi$, $\mu \to -\mu$, fold and tip $\theta \to \pi - \theta$ of the Kohn-Sham theorem T3). Gauge-anomaly cancellation of the four-dimensional zero-mode theory is that of one Standard Model generation (STANDARD); anomalies of the eight-dimensional bulk theory are OPEN.
- The Higgs field is ABSENT. Electroweak symmetry is broken at the tip by a boundary condition: the tip carries a NON-LINEAR sigma field $\Sigma \in SU(2)$, $\Sigma^\dagger\Sigma = 1$, transforming as

$\Sigma \to U_L \Sigma U_Y^\dagger$ (INVENTED for this geometry; the structure is that of the electroweak chiral Lagrangian and of Higgsless boundary conditions, STANDARD). Σ has no radial mode, hence no Higgs particle; in the unitary gauge $\Sigma = 1$.

$$S_{tip} = \int_{y=-L} d^4x\, d^3x_t \sqrt{|g_{tip}|}\Big[\frac{v_\Sigma^2}{4}\mathrm{tr}(D_\mu \Sigma^\dagger D^\mu \Sigma) - (\bar{L}\,\Sigma\,\Lambda_\ell\,\ell_R + \bar{Q}\,\Sigma\,\Lambda_q\,q_R + \mathrm{h.c.})\Big]$$

with the right-handed doublets $\ell_R = (\nu_R, e_R)$, $q_R = (u_R, d_R)$ and the diagonal coupling matrices $\Lambda_\ell = \mathrm{diag}(\lambda_\nu, \lambda_e)$, $\Lambda_q = \mathrm{diag}(\lambda_u, \lambda_d)$ (the Yukawa structure of the electroweak chiral Lagrangian, gauge invariant because $U(1)_Y$ acts on Σ from the right). The first term gives the W and Z their masses with $m_Z = m_W/\cos\theta_W$ at tree level (custodial $SU(2)$ of Σ); the second gives the fermion masses through the values of the fermion profiles at the tip.

- QCD (STANDARD): the gluons are the massless x_0 zero modes of G_M^α; below Λ_{QCD} the u and d quarks are confined into protons (uud) and neutrons (udd), whose masses come from QCD and are inputs of the cosmology (section 7).

4.4 The mass mechanism (the author's idea, made precise; DERIVED in `FIELD_EQUATIONS.md` section 4)

Every field is expanded on the cell, $\Phi(x) = \sum_n \phi_n(x_1, \ldots, x_4)\, u_n(x_0)$, with u_n the eigenfunctions of its x_0 operator (periodic with period ℓ_0, the fold and tip conditions). The four-dimensional mass is the eigenvalue, $m_n^2 = \lambda_n$, and the Rayleigh quotient gives

$$m^2 = \frac{\int dy\, w(y)\, |\partial_y u|^2/f^2 + \ldots}{\int dy\, w(y)\, |u|^2}, \qquad w(y) = e^{3Hy} \text{ (the measure } \sqrt{|g|}\, dx_0 \propto e^{3Hy} dy).$$

For a profile localised within a proper length $\Delta s = f\,\Delta y$ the uncertainty relation on the cell gives $\langle p_{x_0}^2 \rangle \geq \hbar^2/(4\Delta s^2)$, hence

$$m\,c^2 \geq \frac{\hbar c}{2\,\Delta s} \quad \text{(the author's estimate; exact for the Gaussian profile).}$$

- MASSLESS (no x_0 dependence, or an exact zero eigenvalue): the graviton, the photon and the gluons (flat profiles of unbroken gauge generators), the fold-band quanta of Ψ (the fold zero modes, $\varepsilon = e^{-a_4}|k|$).
- MASSIVE: $W^\pm$ and Z (their profiles bend at the tip, where Σ acts), e, ν, u, d (chiral zero modes localised by μ_ψ, lifted by λ_f at the tip: $m_f = \lambda_f\, \zeta_{fL}\zeta_{fR}$ with $\zeta(\mu)^2 = 2f\mu\, e^{-2f\mu L}/(1 - e^{-2f\mu L})$ the normalised value of the zero mode ($e^{f\mu y}, 0$) at the tip), the bulk-band quanta of Ψ and every excited x_0 mode (the Kaluza-Klein tower, $m_n \simeq n\pi/(fL)$ for $n \geq 1$).
- Because the profiles depend exponentially on $\mu_\psi f L$, order-one parameters produce the observed hierarchy $m_\nu \ll m_e \ll m_u, m_d \ll m_W$ (DERIVED as a fit in `SOLUTION.md`).
- Every x_0-generated mass depends on the block size f: $m \propto 1/f$ for Kaluza-Klein masses and $m \propto \exp(-\mu f L)$ factors for the localised fermions. A deflating block ($h_4 < 0$) therefore makes masses drift in time; the model requires $|h_4| \ll \dot{a}_4$ after the primordial era (section 7; the bound is computed in `SOLUTION.md`). The extra times keep deflating (never static), but slowly.

4.5 The portal interactions of Ψ with the Standard Model (INVENTED; gauge invariant)

Ψ is neutral, so every interaction with the Standard Model is through gauge-invariant composite operators. The model contains the lowest ones (eight-dimensional coefficients; the four-dimensional couplings follow from the overlaps of the x_0 profiles):

	operator	**couples Ψ to**	**cosmological role**
P1	$\frac{c_V}{\Lambda_D^6}\,(\bar{\Psi}\gamma^M\Psi)\,(\bar{\psi}\gamma_M\psi)$, $\psi \in \{L, e_R, Q, u_R, d_R\}$	e, ν, u, d (vector currents)	dark matter - baryon and - neutrino scattering
P2	$\frac{c_S}{\Lambda_D^6}\,(\bar{\Psi}\Psi)\,(\bar{\psi}_L\Sigma\psi_R + \text{h.c.})\big\vert_{tip}$	e, u, d (scalar densities)	spin-independent dark matter - nucleon scattering
P3	$\frac{\mu_D}{\Lambda_D^3}\,\bar{\Psi}S^{MN}\Psi\,B_{MN}$	photon and Z ($B = \cos\theta_W A - \sin\theta_W Z$)	dark dipole: dark matter - photon drag, Z decays
P4	$\frac{c_Z}{\Lambda_D^4}\,(\bar{\Psi}\gamma^M\Psi)\,\frac{iv_\Sigma^2}{2}\mathrm{tr}\big(\Sigma^\dagger D_M\Sigma\,\tau^3\big)\big\vert_{tip}$	Z (directly)	Z-portal
P5	$\frac{c_D}{\Lambda_D^7}\,(\bar{\Psi}\Psi)\,(\bar{\psi}\,i\gamma^M D_M\psi + \text{h.c.})$	$W^\pm$, Z, γ, gluons with e, ν, u, d	charged-current contact vertices

- Charge conservation forbids any gauge-invariant operator in which the neutral Ψ couples to a single $W^\pm$ without a charged pair of Standard-Model fermions; the $W^\pm$ enter the portal through the covariant derivative of P5 (vertices $\bar{\Psi}\Psi\,W^+\bar{e}\nu$, $\bar{\Psi}\Psi\,W^+\bar{d}u$) and through the gauge self-interactions (DERIVED in `FIELD_EQUATIONS.md` section 5).
- The coefficients are free parameters bounded by cosmology and by laboratory limits (`SOLUTION.md` section 6). The word CRITICAL of the order is implemented as follows: these are the only couplings through which the dark sector exchanges energy and momentum with the visible sector; without them Ψ interacts with the Standard Model only through gravity.

5. Dimensional reduction (DERIVED at leading order in `FIELD_EQUATIONS.md`)

- The four-dimensional theory is obtained by integrating over the cell and the extra-time block with the zero-mode profiles. The reduced Planck mass and gauge couplings carry the proper volume $V_\perp = 2\int_{-L}^{0} f\,e^{3Hy}\,dy \times f^3 v_t = 2f^4 v_t(1 - e^{-3HL})/(3H)$ (2 for the two halves of a cell, v_t the coordinate volume of the extra times; their compactness is ASSUMED, normalisation A of the dark-sector record): $M_{Pl}^2 = V_\perp/\kappa$, $g^2 = \hat{g}^2/V_\perp$.
- The extra times are TIME-LIKE. Modes with extra-time momentum q have $\omega^2 = M^2 + k^2 - q^2$ and are tachyonic for $q > M$ (DERIVED in the dark-sector record); the model keeps only the good sector $q = 0$ (ASSUMED), as the Kohn-Sham record does. The stability of the full theory is OPEN.

6. The cosmological inventory

component	field	x_0 mode	equation of state
photons (CMB)	A_μ zero mode	flat, massless	$w = 1/3$
neutrinos (CNB)	ν_L zero mode	localised, $m_\nu \lesssim 0.1$ eV	relativistic, then non-relativistic
baryons	$u, d \to p, n$	localised, QCD	$w = 0$
electrons	e	localised	$w = 0$ after $e^\pm$ annihilation
dark matter	Ψ bulk band	massive, $\varepsilon \geq m$	$w \simeq 0$ when cold and f frozen
dark radiation	Ψ fold band	massless zero modes	$w = 1/3$ (ΔN_{eff})
dark condensate	$\bar{\Psi}\Psi$	homogeneous	$w_c = \lambda S/(2m + \lambda S)$, drifting
hidden curvature	geometry, $6H^2k$, $k = e^{2a_4}/b_4$	-	acts like a cosmological term when k is constant

7. What the model predicts and what it requires (statements to be computed)

- The background equations (DERIVED from the record, `FIELD_EQUATIONS.md` section 2): in Einstein gravity the x_4 constraint is $3\dot{a}_4^2 + 12\,\dot{a}_4 h_4 + 6h_4^2 - 6H^2k = \kappa\rho + \Lambda$ with $k = e^{2a_4}/b_4$: a Friedmann equation with the extra term $12\dot{a}_4 h_4 + 6h_4^2$ of the deflating block and the hidden-curvature term $6H^2k$, which is constant when the block is static and acts like a cosmological constant.
- The masses: the W, Z and fermion masses are fitted with order-one parameters; the photon, gluon and graviton are exactly massless (`SOLUTION.md` section 4).
- CMB, CNB and BAO: computed in the four-dimensional effective theory at order ϵ (`SOLUTION.md` section 5): the neutrino temperature $T_\nu = (4/11)^{1/3} T_\gamma$ (STANDARD, unchanged as long as Ψ does not heat the neutrinos), the sound horizon at the drag epoch, the angular acoustic scale and the BAO scale, and the bounds that the deflation rate, the dark radiation and the portal couplings must respect.

8. Assumptions and open problems (honesty rule)

- ASSUMED: the regular tip at $y = -L$; the compact extra times (normalisation A) and the good sector $q = 0$; the kink bulk masses; the tip sigma field; the portal operators; one generation; the zero-mode truncation at order 1; (1.1) the block stabiliser and the exact fold-reflection parity that keeps the odd-parity dark matter stable.
- CHANGED (1.1): $w_3 = 1/768$ and $\Lambda = -24H^2 + \delta\Lambda$ (section 4.1). The solution also finds what the model requires of the data: $m_{KK} \approx 121$ TeV from ρ_0, and, with one generation, a

decoupling of the Ψ fold band inside the QCD crossover for $N_{eff} \approx 3$ (`SOLUTION.md` sections 4 and 6).

- OPEN: the stability of a theory with three time-like extra dimensions (ghosts and tachyons beyond the good sector); perturbative unitarity of WW scattering without a Higgs (it holds up to the Kaluza-Klein scale $\sim \pi/(fL)$ only if that scale lies below about $4\pi v_\Sigma \approx 3$ TeV, as in all Higgsless models); a dynamical mechanism that slows the deflation of the block after the primordial era; the origin of the second and third generations; the derivation of the portal coefficients; the non-adiabatic evolution of Ψ.
- Consistency with QCD and the electroweak theory is at the level of the gauge group, the representations, anomaly cancellation and the electroweak chiral Lagrangian (no Higgs); precision electroweak constraints on Higgsless models (the S parameter) apply and are OPEN for this geometry.